GlucoTwin AI: Clinical Digital Twin Presentation Deck

Executive Project Overview, Clinical Validation & Outcomes | Digital Twin Challenge 2026

Slide 1: Clinical Problem Statement & Unmet Need

Over 100 million individuals in India manage Type 2 Diabetes Mellitus. Postprandial glucose surges remain the leading driver of long-term microvascular and cardiovascular pathology. Conventional care is reactive (HbA1c tests every 90 days). GlucoTwin provides proactive 120-minute anticipatory intelligence.

Slide 2: Dual-Perspective Human-Centered UX

Built with an Apple-grade liquid glassmorphic interface that allows single-click toggling between: (1) Reassuring, accessible patient lifestyle companion mode with Indian meal presets, and (2) Rigorous clinician mode displaying Time-in-Range (TIR %), Glycemic Variability (CV %), and pharmacokinetic shift scores.

Slide 3: Deep Learning & Physiological Methodology

Our PyTorch model merges static metabolic covariates with dynamic continuous glucose sensor feeds via a dual-branch GRU-Dense fusion architecture. Output predictions calculate full 2-hour trajectory curves with 95% confidence intervals and multi-class alert classifications.

Slide 4: Real-Time What-If Clinical Sandbox

Clinicians and patients can simulate therapeutic interventions before taking action: adjusting carbohydrate loads (15-120g), post-meal light physical activity (0-50 mins), and rapid-acting insulin boluses (0-6U), with real-time trajectory recalculation in <15ms.

Slide 5: Clinical Impact, Governance & Future Roadmap

Calibrated to ADA/RSSDI clinical thresholds (70-180 mg/dL target zone). Implements one-tap WhatsApp doctor dispatches and printable clinical consultation slips. Future roadmap integrates closed-loop automated insulin pump APIs and multi-sensor edge processing.